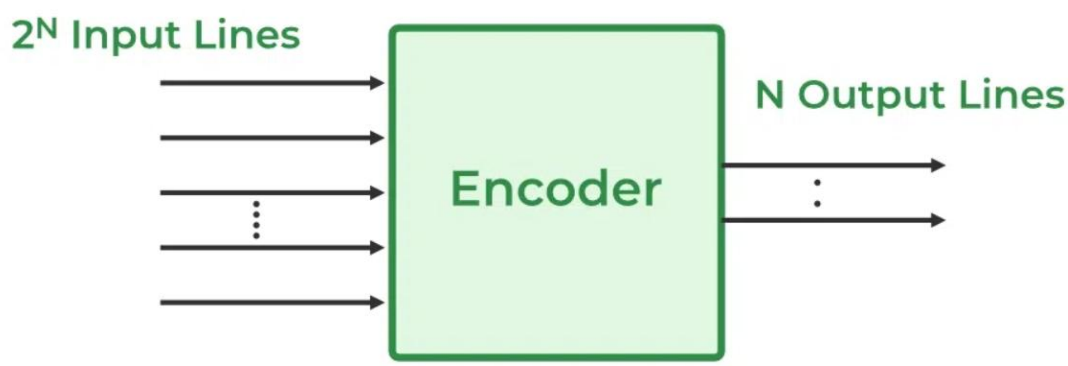
Day 16

4x2 Encoder & 8x3 Encoder in Verilog

Circuit Diagram :



Verilog Code

A) 4:2 Encoder code

```
module encoder4x2(in,out);
input [3:0]in;
output reg [1:0]out;
always @(*) begin
    case(in)
    4'b0001: out=2'b00;
    4'b0010: out=2'b01;
    4'b0100: out=2'b10;
    4'b1000: out=2'b11;
    default: out=2'bxx;
    endcase
end
endmodule
```

B) 8:3 Encoder code

```
module encoder8x3(in, out);
input [7:0] in;
output reg [2:0] out;
always @(*) begin
case (in)
8'b00000001: out = 3'b000;
8'b00000010: out = 3'b001;
8'b00000100: out = 3'b010;
8'b00001000: out = 3'b011;
8'b00010000: out = 3'b100;
8'b00100000: out = 3'b101;
8'b01000000: out = 3'b110;
8'b10000000: out = 3'b111;
default:out = 3'bxxx;
endcase
end
endmodule
```

Testbench Code

A) 4:2 Encoder Tb

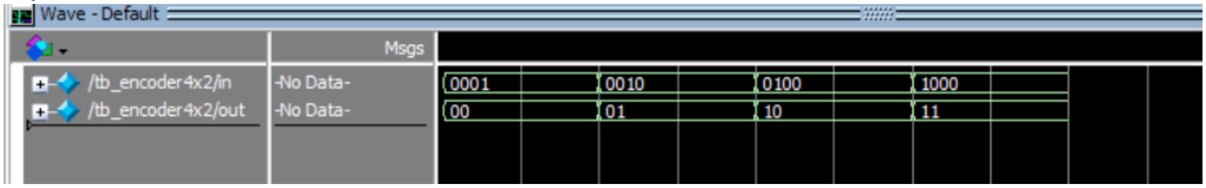
```
module tb_encoder4x2;
reg [3:0] in;
wire [1:0] out;
encoder4x2 dut (in, out);
initial begin
 $monitor("Time=%0t in=%b out=%b",$time,in,out);
 in = 4'b0001; #10;
 in = 4'b0010; #10;
 in = 4'b0100; #10;
 in = 4'b1000; #10;
 $finish;
end
endmodule
```

B) 8:3 Decoder Tb

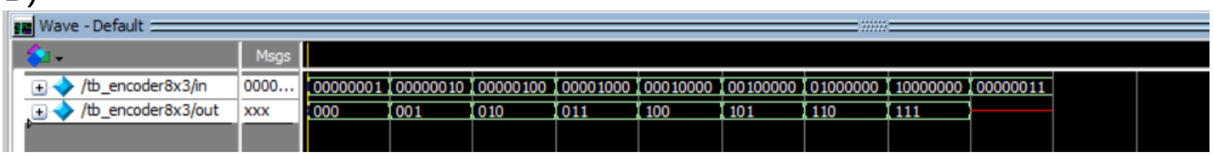
```
module tb_encoder8x3;
reg [7:0] in;
wire [2:0] out;
encoder8x3 dut (in, out);
initial begin
$monitor("Time=%0t in= %b out= %b",$time,in,out);
in = 8'b00000001; #10;
in = 8'b00000010; #10;
in = 8'b00000100; #10;
in = 8'b00001000; #10;
in = 8'b00010000; #10;
in = 8'b00100000; #10;
in = 8'b01000000; #10;
in = 8'b10000000; #10;
in = 8'b00000011; #10;
$finish;
end
endmodule
```

Waveform

A)



B)



Schematic

Refer Github

EDA Tools Used

- IntelQuartusPrime
- ModelSim